

In situ structure and dynamics of DNA origami determined through molecular dynamics simulations

Jejoong Yoo^a and Aleksei Aksimentiev^{a,b,1}

^aDepartment of Physics and Center for the Physics of Living Cells and ^bBeckman Institute for Advanced Science and Technology, University of Illinois at Urbana-Champaign, Urbana, IL 61801

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The DNA origami method permits folding of long single-stranded DNA into complex 3D structures with subnanometer precision. Transmission electron microscopy, atomic force microscopy, and recently cryo-EM tomography have been used to characterize the properties of such DNA origami objects, however their microscopic structures and dynamics have remained unknown. Here, we report the results of all-atom molecular dynamics simulations that characterized the structural and mechanical properties of DNA origami objects in unprecedented microscopic detail. When simulated in an aqueous environment, the structures of DNA origami objects depart from their idealized targets as a result of steric, electrostatic, and solvent-mediated forces. Whereas the global structural features of such relaxed conformations conform to the target designs, local deformations are abundant and vary in magnitude along the structures. In contrast to their free-solution conformation, the Holliday junctions in the DNA origami structures adopt a left-handed antiparallel conformation. We find the DNA origami structures undergo considerable temporal fluctuations on both local and global scales. Analysis of such structural fluctuations reveals the local mechanical properties of the DNA origami objects. The lattice type of the structures considerably affects global mechanical properties such as bending rigidity. Our study demonstrates the potential of all-atom molecular dynamics simulations to play a considerable role in future development of the DNA origami field by providing accurate, quantitative assessment of local and global structural and mechanical properties of DNA origami objects.

nucleic acids | self-assembly | nanotechnology | nanopore

Self-assembly of DNA into complex 3D objects has emerged as a new paradigm for practical nanotechnology (1, 2). Among many methods that have been put forward to use self-assembly of DNA (2), DNA origami (3) stands out through its conceptual simplicity and infinite range of possible applications (1, 2). The basic principle of the method is folding of a long (tens of thousands of nucleotides) DNA strand into custom 2D or 3D shapes using short oligonucleotides (“staples”) (3). Since its first demonstration in 2006, the DNA origami method has been used to self-assemble complex 3D objects with subnanometer precision (4) that can serve as static structures (1, 2, 5, 6), and also perform active functions (7–10). Recent methodological advances (11) have made practical applications (11–14) of DNA origami feasible.

Due to the intrinsic complexity of DNA origami, computational tools have been essential for the development of the field. In the seminal work, Rothemund used a custom computer code to design sets of staple strands to fold the M13 viral genome into unique 2D patterns (3). Design of 3D origami has been facilitated by the caDNAno program (15), which can semiautomatically generate a set of staple strands to realize folding of the M13 genome into a user-defined 3D object. The structural stability of caDNAno designs can be assessed using the CanDo (16) program, which provides instant review of the caDNAno outputs based on a continuum mechanics approximation. However, the computational models of DNA origami objects have not yet taken explicitly into account the atomic-scale interactions that govern self-assembly of DNA. As pairing of complementary DNA strands into a DNA

duplex is a balancing act involving electrostatic, hydrophobic, and solvation forces (17), one can expect all-atom modeling to provide the most comprehensive description of DNA origami, especially under environmental stress.

Until recently, experimental characterization of DNA origami was limited to atomic force spectroscopy (8), small-angle X-ray scattering (8), and transmission electron microscopy (4, 5). Recently, superresolution optical imaging (10), FRET (13), and magnetic tweezer (18) have been applied to DNA origami objects to infer information about their in situ structure and dynamics. The only atomic-level model of DNA origami in situ has been derived from cryo-electron microscopy (cryo-EM) (19), which revealed considerable deviations of the in situ structure from the idealized design. Despite the tremendous insights brought about by the cryo-EM structure, the structural dynamics and local mechanical properties of DNA origami have remained unknown. Furthermore, the complexity of the cryo-EM reconstruction makes this method unlikely to be routinely used to characterize DNA origami objects.

Here, we report all-atom molecular dynamics (MD) simulations of DNA origami structures. By following trajectories of all of the atoms comprising a DNA origami object and the surrounding solution, we could characterize their structure and dynamics in microscopic detail. Below, we describe the in situ structures of several DNA origami objects, their temporal fluctuations, the conformations of key structural elements, and local mechanical properties. While we find the global properties of the simulated DNA origami objects to be in good agreement with experimental measurements, our simulations reveal the intrinsically dynamic nature of DNA origami objects.

Significance

Folding of DNA strands into complex three-dimensional nanoscale objects—DNA origami—has emerged as a new paradigm for practical nanotechnology. Although computer-aided design tools are often used to illustrate the idealized shape of such objects, their structural behavior in solution has been poorly characterized. Using a large supercomputer system, we created trajectories of millions of atoms to obtain the first atomically precise portrait of DNA origami in solution. The DNA origami objects were found to undergo considerable structural fluctuations, significantly departing from their idealized conformations at the nanometer scale. Analysis of the fluctuations provided a detailed map of local mechanical properties of DNA origami. Our work demonstrates the potential for atomistic simulations to facilitate rational engineering of DNA origami objects.

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¹To whom correspondence should be addressed. E-mail: aksiment@illinois.edu.

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Results and Discussion

To enable direct comparison between simulation and experiments, we designed three DNA origami structures following standard design principles (5). The first two structures were rod-like monoliths based on the honeycomb (HC) (5) and square (SQ) (20) DNA lattices. The third structure had a programmed 90° bend (HC-90°). Fig. 1 *A–C* illustrates our target designs.

We used caDNAno to obtain an approximate spatial model of each DNA origami object and to design DNA strands that could realize the spatial model. The caDNAno models were converted

to an all-atom representation using a homemade program. The all-atom structures were merged with solvent and simulated using the MD method as described in *Materials and Methods*. A detailed description of our caDNAno designs and the conversion procedures is provided in *SI Appendix* and *SI Appendix*, Figs. S1–S3. *SI Appendix*, Table S1 provides a summary of the simulations performed.

Structural Stability of the All-Atom Models. During our MD simulations, the three DNA origami objects maintained their overall

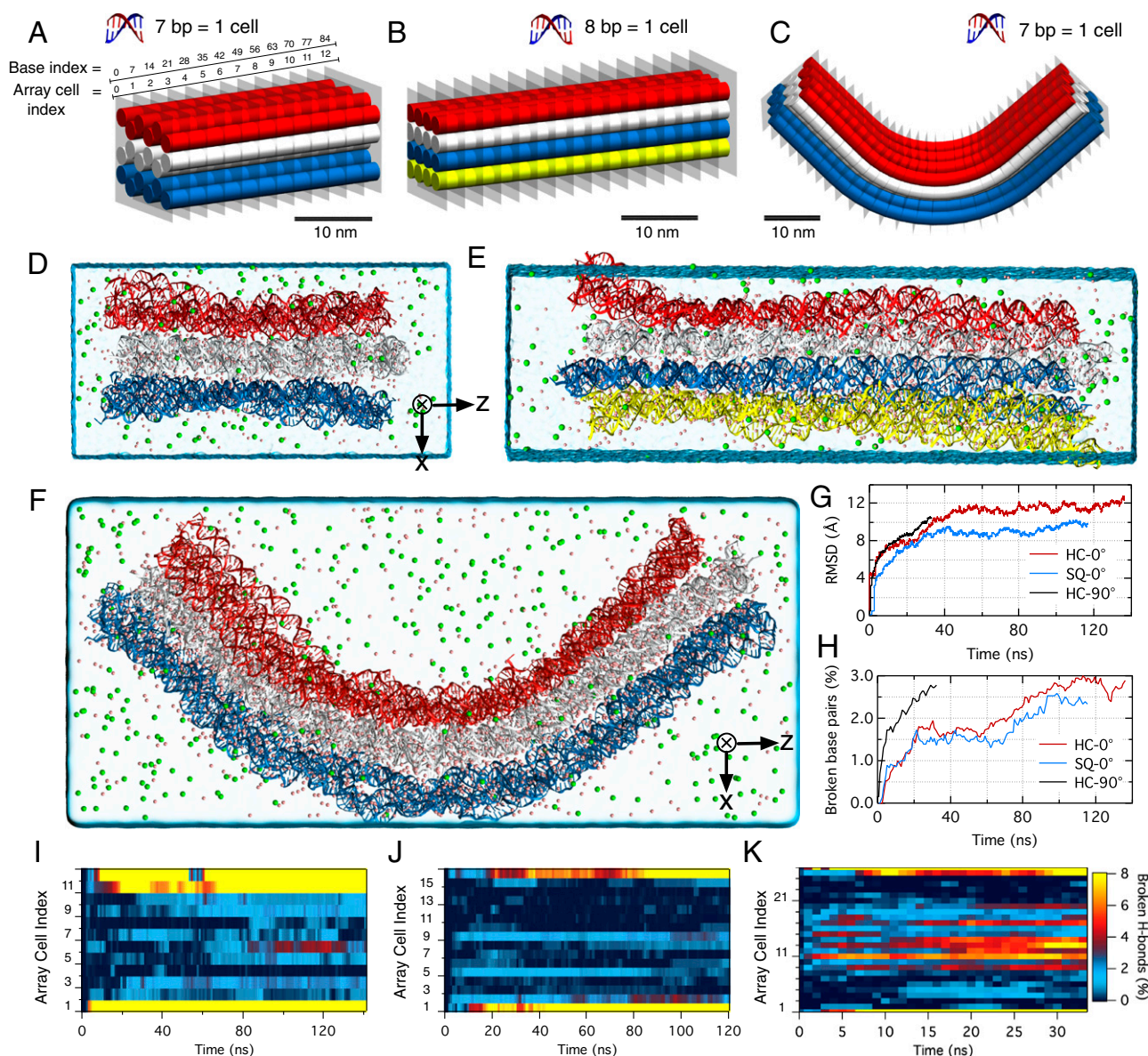


Fig. 1. All-atom MD simulations of DNA origami structures. (*A–C*) Schematic representations of the three DNA origami structures considered in this work: (*A*) straight 6-by-3 HC-pleated structure, (*B*) straight 4-by-4 SQ-pleated structure, and (*C*) 6-by-3 HC-pleated structure having a 90° programmed bend (HC-90°). Each cylinder represents one DNA duplex; all cylinders within the same layer of a 3D structure have the same color. The base index (*A*) indicates the locations of individual base pairs within each cylinder. The cylinders (duplex DNA) are connected to one another through crossovers (HJs) every seven (HC or HC-90°) or eight (SQ) base pairs in each cylinder (5). Gray semitransparent planes in *A–C* indicate where such crossovers occur. The crossover planes naturally divide each origami structure into an array of 7 or 8 bp cells; the array cell index specifies the position of an array cell in each cylinder. (*D–F*) The conformations of the HC (*D*), SQ (*E*), and HC-90° (*F*) structures at the end of our production MD runs. Layers of DNA in the origami structures are colored as in *A–C*. Mg^{2+} and Cl^{-} ions are shown as pink and green spheres, respectively; water is shown as a semitransparent molecular surface. The systems contain 802,149 (HC), 943,837 (SQ), and 2,799,156 (HC-90°) atoms. (*G*) Root-mean-square deviation (rmsd) of the DNA atoms from their initial coordinates during the MD simulations. (*H*) The fraction of base pairs broken during the MD simulations. A base pair is considered broken when the distance between the N1 atom of the purine base and the N3 atom of the pyrimidine base exceeds 4 Å. (*I–K*) The fraction of broken base pairs in the individual array cells of the HC (*I*), SQ (*J*), and HC-90° (*K*) structures versus simulation time.

shape. Fig. 1*A–F* provides one-to-one comparisons of the target designs and the instantaneous microscopic conformations of the three objects observed at the end of our production runs. Fig. 1*G* shows the root-mean-square deviation (rmsd) of the DNA atoms from their initial coordinates. The rmsd values increase initially but reach a plateau after ~40 ns for the HC and SQ systems. Because of its excessively large size (~3 million atoms), the HC-90° system was simulated for a considerably shorter time interval (~35 ns). However, we expect the HC-90° system to have reached an equilibrium conformation at the end of the production run as the rmsd of the HC and SQ systems reached a plateau after 40 ns. Fig. 1*H* shows the fraction of broken base pairs during the MD runs. In all three systems, the fraction was observed to slowly increase with time, but remained less than 3%.

During our production runs, the DNA origami objects maintained their straight (HC and SQ) and bent (HC-90°) conformations (*SI Appendix, Fig. S4 A and B*). In the central portions of the HC and SQ structures, the DNA helices were found to closely follow their prescribed arrangement of an HC or SQ lattice (*SI Appendix, Fig. S4 C and D*). Consistent with experimental estimations (5), the average interhelical distance in the lattices was about 23–24 Å.

Fig. 1*I–K* characterizes the local structural integrity of the DNA origami objects by showing how the fraction of broken base pairs in the individual array cells of the objects (defined in Fig. 1*A–C*) changes during our MD runs. Peripheral regions of the DNA origami objects were found to exhibit a considerably greater fraction of broken base pairs (>5%) than the internal regions, where the fraction remained <2%.

Overall, the local structure of the internal regions of the straight DNA origami objects (HC and SQ) remained stable; the fraction of broken base pairs fluctuated around 1% (Fig. 1*I* and *J*). However, our simulations of the HC and SQ systems indicate that large reversible deformations of the local structure can occur on the time scale of hundreds of nanoseconds. For example, the fraction of broken base pairs in array cell 6 of the HC structure reached 4–5% at ~100 ns (Fig. 1*I*). Unlike peripheral regions of the DNA origami structures, the base pairs forming array cell 6 were constrained by the surrounding duplex DNA, and therefore, the local structural defect could be repaired within the time scale of our simulation (140 ns). *Movies S1–S3* illustrate the dynamics of base pairing at the single-nucleotide level for individual layers of the HC structure. Among the six base pairs from the internal regions that remained broken for 10 ns or longer, five located at the nicks in duplex DNA.

In the case of the HC-90° structure, the fraction of broken base pairs within the programmed bend region (array cell index 7–20) reached ~1–8% after ~15 ns (Fig. 1*K*). Although the number of insertions and deletions was designed to minimize the structural stress introduced by the 90° bend, some stresses remained in the structure. Because helical turn per array cell should be 240° regardless of insertions or deletions, one turn per base pair can range from ~27° (two insertions per cell) to 48° (two deletions per cell) in the bend region. Thus, the bend region could be expected to have a less ordered structure than fragments of straight DNA origami. Indeed, Sobczak et al. (11) reported difficulties with folding a seven-layer HC design having a sharp programmed bend.

SI Appendix, Fig. S5 characterizes the distributions of Mg^{2+} in the HC system during the production simulation. The charges of Mg^{2+} ions completely neutralize the charge of DNA inside the origami (*SI Appendix, Fig. S5A*). The ions are located, on average, 5.4 Å away from the phosphorous atoms of the DNA backbone (*SI Appendix, Fig. S5B*). The diffusion coefficients of Mg^{2+} inside HC and SQ structures are 0.15 and 0.12 10^{-5} cm²/s, respectively, which is considerably lower than 0.52×10^{-5} cm²/s that we measured in 50 mM MgCl_2 bulk solution.

Conformational Dynamics of the DNA Origami Objects. During our MD simulations, the DNA origami objects underwent structural transformations that could be described as initial relaxation of the idealized structural models (first ~40 ns; Fig. 1*G*) and subsequent fluctuations about relaxed equilibrium conformations. To visualize the conformational dynamics in further details, we illustrate several instantaneous configurations of the HC, SQ, and HC-90° systems in Fig. 2*A–C* using a custom “chicken-wire” molecular graphics representation. *Movies S4–S6* illustrate the corresponding MD trajectories.

During the initial relaxation, the DNA origami objects developed unique blue-and-red patterns (Fig. 2*A–C*). The blue color indicates close DNA–DNA contacts of 20 Å or less produced by the crossovers; the red indicates larger than average distance between DNA caused by the electrostatic repulsion. Overall, the DNA–DNA distance was found to range between 18 and 30 Å (Fig. 2*D*), although occasional excursions below and beyond this range could occur because of local disruption of the base pairing order or axial shift of one duplex DNA relative to the other.

After the DNA origami structures had attained their relaxed conformations, the color patterns remained mostly stable, indicating a limited range of thermal fluctuations. To quantitatively characterize the extent of thermal fluctuation, we computed a root-mean-square fluctuation (rmsf) of the DNA base pairs’ centers of mass as a function of their array cell index using the last 60 (HC and SQ) or 10 (HC-90°) ns of the respective trajectories (Fig. 2*E*). With the exception of the terminal array cells, the rmsf values were less than 4 Å. Such fluctuations are expected to play a functional role in many types of DNA nanostructures, for example, by modulating the effective diameter of DNA nanochannels (12, 21, 22), affecting localization of dyes attached to DNA nanoantennas (13) and for superresolution imaging (10), and modulation of permeability of drug-like molecules through DNA nanoboxes (8).

Fig. 2*F* characterizes the local cross-sectional area computed using the last 60 (HC and SQ) or 10 (HC-90°) ns of the respective trajectories. For all systems, the total area in the central regions of the origami objects ranged from 50 to 60 nm². For comparison, the total area of four regular hexagons or nine regular squares each 24 Å on side are ~60 and ~52 nm², respectively. Thus, the DNA helices remain rather tightly packed with the average separation of ~24 Å, consistent with the experimental estimations (5). Total cross-sectional area was found to significantly increase at the ends of the structures. The end effects become negligible about five array cells from the ends.

Structure of the Holliday Junctions. Holliday junctions (HJs)—cross-like assemblies of four DNA strands—are the linchpins of DNA origami structures. HJs are found in DNA origami at each and every staple crossover, where two staple strands bridge two fragments of a scaffold strand (Fig. 3*A*). Single-molecule FRET experiments have shown that, in solution ($[\text{MgCl}_2] \sim 10$ mM) and under no mechanical stress, HJs are likely to form a stacked right-handed conformation characterized by a ~60° interaxial angle due to electrostatic repulsions between DNA duplexes (23, 24). However, adopting such a conformation within a DNA origami is not possible because three or four neighboring DNA duplexes surround each duplex. The action of steric and electrostatic interactions results in local bending of DNA at the HJs (Fig. 2*A–C*).

To quantitatively characterize the conformations of HJs inside our DNA origami structures, we considered center-of-mass coordinates of eight base pairs near each junction (Fig. 3*A*). Fig. 3*B* shows the distribution of the intra- and interhelical distances between the four base pairs nearest to the junction (j, k, j', k') averaged over all staple crossovers and the MD trajectory of the HC structure. The intrahelical distances ($j-k$ and $j'-k'$) peak at around 4.4 Å, which is 1 Å larger than the mean rise per base

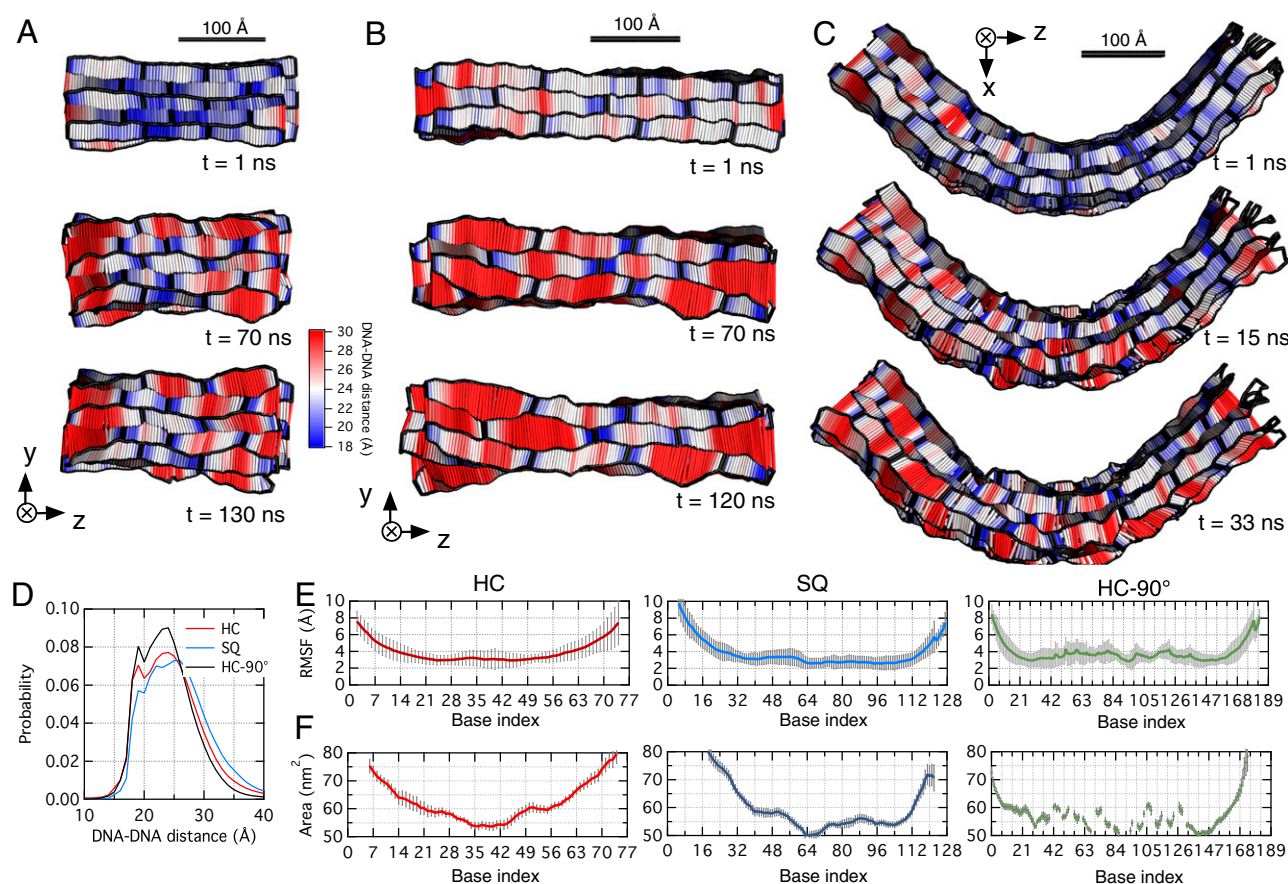


Fig. 2. Conformational dynamics of DNA origami. (A–C) Representative conformations of the HC (A), SQ (B), and HC-90° (C) systems during the MD simulations. The DNA origami structures are shown using a custom color-coded chicken-wire representation. The wire frame (black lines) connects the centers of mass of the DNA base pairs that form continuous double-stranded DNA cylinders or crossovers of the original DNA origami designs. The lines between the centers of mass of the same index base pairs connect the wire frame. The length of the lines indicates the local inter-DNA distance, which is color-coded. [Movies S4–S6](#) and [SI Appendix, Figs. S6 and S7](#) illustrate these simulation trajectories. (D) The distribution of local inter-DNA distances in the three DNA origami structures. (E) Local rmsf of the DNA origami structures. The rmsf values were computed using the coordinates of the base pairs' centers of mass and averaged over last 60 (HC and SQ) or 10 (HC-90°) ns of the corresponding MD trajectory. The error bars indicate the SDs of the rmsf values computed over 18 (HC) or 16 (SQ) helices of a given base index. (F) The local cross-sectional area of the DNA origami structures. The error bar at each base index indicates the SD in the time series data with ~ 5 ps time step.

pair of a canonical B-DNA duplex. Thus, HJs in DNA origami appear to be under mechanical stress. The mean interhelical distance at the junction ($j-k'$ and $k-j'$) is 18.6 Å, consistent with the recent cryo-EM data (19). At the junction, each double helix is bent at $\sim 149^\circ$, whereas the angle between the two double helices is close to 24° (Fig. 3C). The distribution of the $i-j-k'-l'$ and $i-j-k'-l'$ dihedral angles has a mean of -4° (Fig. 3D), which indicates that the helices at the junction form a slightly left-handed antiparallel conformation. Such a conformation is a considerable departure from the conformation of free HJs in solution characterized by a right-handedness and $\sim 60^\circ$ dihedral angle (23, 24). In an ideal DNA origami structure, the two DNA helices of the junctions are nearly antiparallel. However, the steric interaction of the two helices (23, 24) causes significant structural distortions of the junction, which manifests itself as an increase of rise per base and bending of the helices.

[SI Appendix, Fig. S8](#) details our analysis of the HJs' conformations in the SQ lattice structure. The type of the DNA origami lattice was found to have negligible effect on the most likely conformations of the junctions.

Mechanical Properties of DNA Origami Structures. Mechanical properties of molecular objects can be determined from the analysis of their structural fluctuations (25, 26). To perform such analysis on

our DNA origami objects, we expressed their all-atom conformations in terms of local deformation tensors. For each DNA origami object, we defined a contour s to span the entire structure and a triad of vectors $\{\hat{t}_i(s)|i=1,2,3\}$ to define local structural order (Fig. 4A). The local deformation tensor was computed from the relative torsion of two neighboring triads (see [SI Appendix](#) for precise definition).

Having defined the triad vectors as a function of s , we used the Frenet description (25, 26) to quantitatively characterize local structural deformations of DNA origami. The change of the triad vectors along the contour, $d\hat{t}_i/ds$, was described using generalized torsion ω_i ($i=1,2,3$) as $d\hat{t}_i/ds = \epsilon_{ijk}\omega_j\hat{t}_k$, where ϵ_{ijk} is the Levi-Civita tensor.

Fig. 4B and C shows the average values of generalized torsions $\omega_{1,2,3}$, for our two monolith designs (HC and SQ). The mean torsions ω_1 , ω_2 , and ω_3 of the HC structure are close to zero, whereas ω_3 of SQ is consistently below zero with the mean of -1.3° . Indeed, in our MD simulations, the SQ system developed a visible overall twist about the z-axis (Fig. 2B and [Movie S5](#)). A similar twist was observed in the cryo-EM structure of a SQ lattice DNA origami object (19). Experimentally, it is possible to correct the twist by introducing a small number of deletions in the structure, which would relax the unnatural “three turns per 32 base pairs” constraints of the SQ lattice (5).

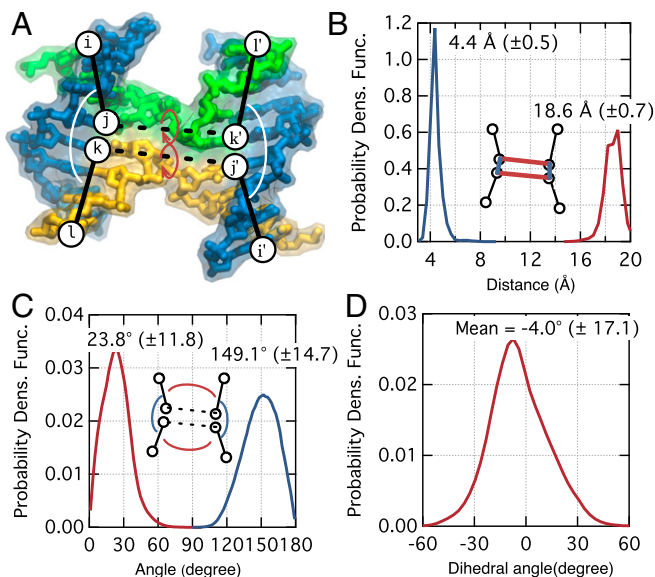


Fig. 3. The structure of HJs in DNA origami. (A) A typical conformation of an HJ in a straight HC structure. The two crossover staple strands are shown in green and yellow; the fragments of the scaffold strand are shown in blue. Points $i, i', j, j', k, k', l, l'$ indicate the locations of the centers of mass of the base pairs at the junction (j, j', k, k') and three base pairs away from the junction (i, i', l, l'). (B) The distributions of distances between base pairs at the junction. Blue corresponds to the intrahelical distance $j-k$ (or $j'-k'$); red corresponds to the interhelical distances $j-k'$ (or $k-j'$). (C) The bending angle of the helices at the junction (blue) and the angle between the helices at the junction (red). (D) The distribution of the dihedral angles $i-j-k'-l'$ and $i'-j'-k-l$. All distributions were obtained by averaging over the last 60 ns of the HC trajectory and over all junctions in the structure. In B–D, the mean and SD of each distribution are listed. Similar conformations of the HJs were observed in our simulations of the SQ lattice origami (SI Appendix, Fig. S8).

The generalized torsion ω_1 in Fig. 4D indicates the local curvature of the HC-90° structure as a function of array index. The local curvature switches from zero to $\sim 8^\circ$ over three array cells (21 base pairs) at the beginning and at the end of the programmed bend region. Over the central 14-array-cell programmed bend region, the curvature is rather evenly distributed. However, subtle variation of the local curvature along the HC-90° structure

is apparent. The local curvature appears to oscillate with a roughly 21-bp periodicity, which is an intrinsic feature of the HC lattice.

Using the Frenet analysis, we could quantify the mechanical rigidity of our straight origami designs. As demonstrated in SI Appendix, Fig. S9 A–C, the distributions of $\omega_{1,2,3}$ can be accurately described using Gaussians. Thus, we can approximate the generalized torsional energy density, $E_i(s)$, as $E_i(s) = k_B T \alpha_i (\omega_i - \omega_{0i})^2 / 2$, where $\alpha_{1,2}$ are the bending moduli and α_3 is the twist modulus (25, 26). The persistence length, l_p , is related to the bending moduli as $1/l_p = 0.5(1/\alpha_1 + 1/\alpha_2)$ (26).

The bending, $\alpha_{1,2}$, and twist, α_3 , moduli as a function of array cell index of all systems are shown in SI Appendix, Fig. S9 D–F, whereas the mean torsional moduli of the HC and SQ systems, averaged over all array cell indices, are shown Fig. 4E. Overall, α_1 and α_2 of the HC and SQ systems are comparable, indicating that they are isotropic for bending. Such isotropy could be expected for the SQ structure because of its symmetric four-by-four design and is reasonable for the HC structure because of the comparable widths of the structure in the directions of $\hat{t}_1$ and $\hat{t}_2$. The twist modulus α_3 is an order of magnitude smaller than $\alpha_{1,2}$ for both HC and SQ systems: $\sim 3 \mu\text{m}$ for both.

Fig. 4F plots the persistence lengths of the HC and SQ systems. Although the cross-section areas of the HC and SQ systems are comparable (Fig. 2F), the HC-pleated design is significantly more rigid than the SQ pleated one: $l_p = 52.8 \pm 17.4 \mu\text{m}$ versus $29.4 \pm 9.8 \mu\text{m}$. Our estimates are in general agreement with experimental measurement of the persistence length of a DNA origami nanotube, $l_p \sim 17 \mu\text{m}$ (27).

Conclusions

We have shown that the all-atom MD method can reproduce experimentally known structural properties of DNA origami objects and also provide information that is currently not accessible to experiment. As cluster computing becomes more commonplace, we expect MD simulations to play an essential role in the development of the DNA origami field. At the very least, an all-atom MD simulation can be used to assess the overall geometry and stability of a DNA origami object with ultimate temporal and spatial resolution. In this regard, the MD method is complementary to less accurate but more computationally efficient methods based on continuum mechanics. For a proven DNA origami design, an all-atom MD simulation can eliminate the need for cryo-EM reconstruction. Furthermore, the MD method can provide complete structural information about

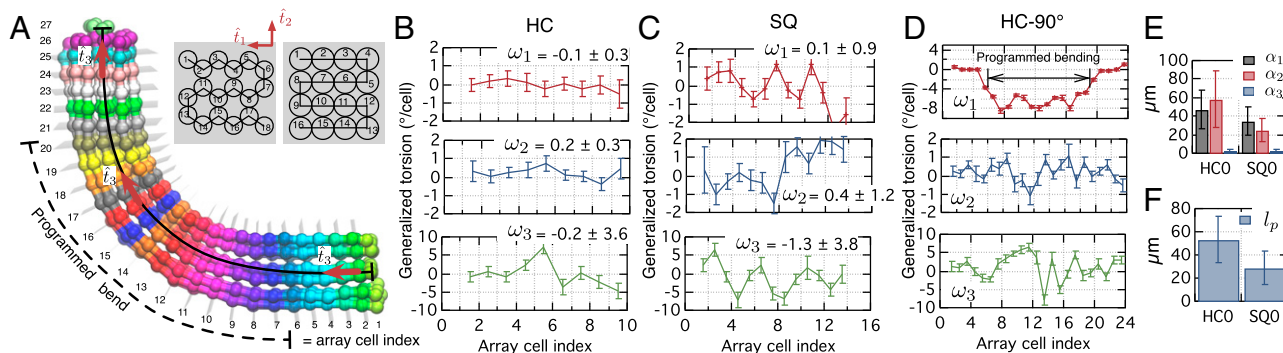


Fig. 4. Mechanical properties of DNA origami structures. (A) Definition of the unit vector triads, $\{\hat{t}_i(s)|i=1,2,3\}$. The HC-90° system is shown using spheres that represent the center of mass (CoM) of each 7-bp cell connected by cylinders; spheres of the same array cell index have the same color. For each array cell, a semitransparent gray plane represents a least-square fit to the CoM of the cells. The contour s is defined to connect the CoM of the array cells (black solid line). For each least-square fit plane, $\hat{t}_3$ is the unit vector tangential to the contour s . The insets illustrate the definitions of $\hat{t}_{1,2}$ for the HC (Left) and SQ (Right) lattices. For the HC and SQ lattices, $\hat{t}_1$ is defined as a unit vector connecting helices 7 and 11 or 5 and 8, respectively, within the least-square plane. $\hat{t}_2 = \hat{t}_3 \times \hat{t}_1$. (B–D) Local generalized torsions $\omega_{1,2,3}$ in the HC (B), SQ (C), and HC-90° (D) objects as a function of array cell index. The error bar for each array cell index indicates the SD of 30 block averages of the corresponding time series data. (E and F) Generalized rigidities $\alpha_{1,2,3}$ (E) and the persistent length (F) of the HC and SQ systems. The mean value and the SD were obtained by averaging over the data from individual array cells (SI Appendix, Fig. S9 D–F).

each and every nucleotide of a DNA origami object, surrounding water and ions, local mechanical stresses and their microscopic origins. Such capability can be of particular value for designs that incorporate active elements at specific locations of a DNA origami object (2, 12, 28). The utility of the MD method can be fully realized in the development of systems where DNA origami is combined with other structural components, such as lipid bilayer membranes (12, 21), nanoparticles (13, 14), or solid-state nanopores (22, 28, 29) and where physical forces are used to actively control the behavior of the assemblies.

Materials and Methods

Using caDNAno, we designed three DNA origami objects and converted the caDNAno designs to all-atom models in vacuum as described in *SI Appendix*. Following the conversion, each model was submerged in aqueous solution containing ~10 mM MgCl₂. We used the genbox program of the gromacs package (30) to randomly place Mg²⁺ and Cl⁻ ions and to add water to our models. The size of the water box was chosen such that a water buffer of 4 nm or more separated the periodic images of DNA origami objects. The final size of the HC and SQ systems was approximately 16.5 × 16.5 × 32.5 and 14.0 × 14.0 × 47.0 nm³, respectively. Before equilibration, the potential energy of each system was minimized using the conjugate gradient method.

Two different protocols were used to equilibrate our straight (HC and SQ) and bent (HC-90°) DNA origami objects. For the straight objects, 10 ns equilibration was performed applying harmonic restraints to all heavy atoms of DNA bases using the spring constant k of 1 kcal/mol/Å². This equilibration process allowed the DNA backbone to relax and Mg²⁺ to diffuse into the DNA origami objects, while preserving the overall integrity of the designs.

For the bent object, we used a multistep equilibration strategy described in *SI Appendix* and illustrated by *Movie S7*.

All MD simulations reported in the main text were performed using the NAMD program (31), periodic boundary conditions, CHARMM36 force field for DNA (32), the modified TIP3P model of water (33), and custom parameters for ions (34). All Mg²⁺ ions were simulated as Mg²⁺-hexahydrates (34). During the equilibration, the structure of hexahydrate was preserved by harmonically restraining ($k = 1$ kcal/mol/Å²) the distance between Mg²⁺ and water oxygen atoms to ~1.9 Å. These restraints prevented irreversible binding of Mg²⁺ to phosphate oxygens of DNA during initial equilibration (34). In our production runs, the restraints were not applied; nevertheless, the hexahydrate structure remained intact. The van der Waals and short-range electrostatic energies were calculated using an 8–10 Å switching scheme. The long-range electrostatic interactions were computed using the particle-mesh Ewald scheme and the grid size of ~1.2 Å (35). The integration time step was 2 fs; 2–2.6 fs multiple timestepping was used (31). Temperature was held constant at 298 K using a Langevin thermostat (31). Pressure was maintained at one bar using the Nosé–Hoover Langevin piston pressure control (36).

We have also performed production simulations using the AMBERbsc0 DNA force field (37). However, we found the resulting structures of DNA origami objects to be in considerable departure from experimental data (5). These simulations are described in *SI Appendix* and *SI Appendix, Fig. S10*.

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